

Size-weighted vs equal-weighted FedAvg under server-side DP

Dirichlet $\alpha = 0.3$, cifar10_cnn, seed 42. Size-weighted is Flower's FedAvg default (weights proportional to client data), while its DP noise assumes equal weights (sensitivity C/n). Equal weights every client $1/n$, so the noise matches the sensitivity.

Stage C frontier (48 clients, 7 targets)

Attack score: share of rounds where the target's shadow loss is lower with it in training (0.5 = no signal), mean over 7 targets. Both compared at round 50: accuracy at round 50 and attack score over rounds 1–50. The equal-weighted runs trained 57 rounds and are truncated to 50.

CIFAR-10

Setting	Accuracy		Attack score	
	Size-weighted	Equal	Size-weighted	Equal
Vanilla	56.3%	53.3%	0.87	0.86
Global-DP 0.0025	53.3%	52.8%	0.80	0.89
Global-DP 0.004	45.8%	43.1%	0.78	0.78
Global-DP 0.00625	27.6%	26.0%	0.72	0.65
Metric-privacy 0.0025	55.2%	53.5%	0.84	0.89
Metric-privacy 0.004	49.6%	48.8%	0.88	0.86
Metric-privacy 0.00625	42.1%	39.6%	0.83	0.84

EuroSAT (32×32)

Setting	Accuracy		Attack score	
	Size-weighted	Equal	Size-weighted	Equal
Vanilla	75.0%	73.3%	0.76	0.85
Global-DP 0.0025	71.0%	70.2%	0.69	0.65
Global-DP 0.004	57.7%	55.6%	0.66	0.61
Global-DP 0.00625	39.2%	35.4%	0.54	0.53
Metric-privacy 0.0025	75.3%	72.3%	0.68	0.68
Metric-privacy 0.004	66.7%	66.4%	0.64	0.71
Metric-privacy 0.00625	46.0%	44.7%	0.56	0.54

Influence correlation (16 clients, all scored)

Spearman ρ between each influence definition and the per-client attack score, within each setting, noise ratio 0.004. With 16 clients, $|\rho|$ below about 0.5 is within chance.

CIFAR-10

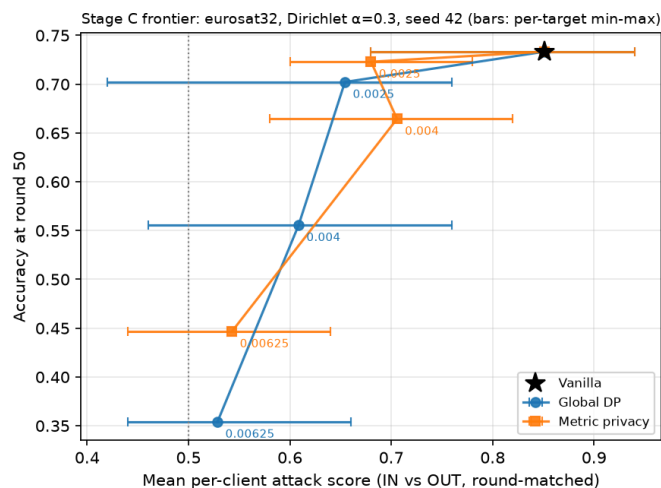
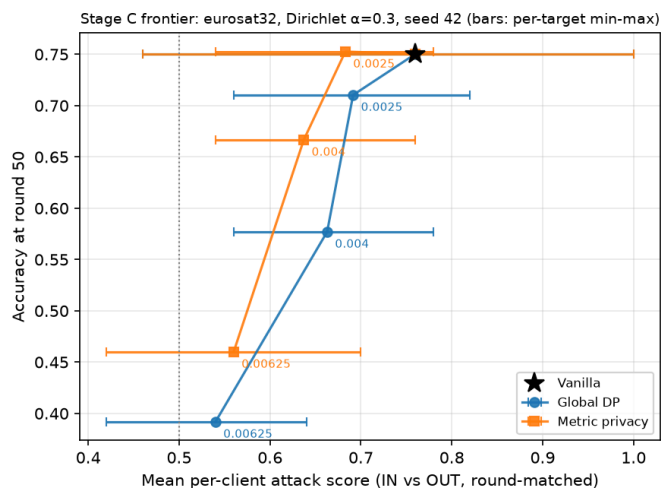
Definition	Size-weighted			Equal		
	Vanilla	GDP	MP	Vanilla	GDP	MP
Client size	+0.80	+0.86	+0.88	-0.28	+0.62	+0.71
Leave-one-out influence	+0.80	+0.86	+0.88	-0.22	+0.65	+0.74
Update size before clipping	+0.81	+0.87	+0.90	-0.25	+0.65	+0.69
Distance from average update	+0.82	+0.89	+0.91	-0.22	+0.65	+0.74
Cosine with average update	+0.74	+0.80	+0.89	-0.04	+0.47	+0.51
How often clipped	—	+0.66	+0.58	—	+0.43	+0.46
Label skew	+0.12	+0.11	-0.07	-0.60	+0.27	+0.32
Training loss	-0.37	-0.22	-0.08	-0.07	+0.05	-0.08

EuroSAT (32×32)

Definition	Size-weighted			Equal		
	Vanilla	GDP	MP	Vanilla	GDP	MP
Client size	+0.82	+0.59	+0.67	+0.38	+0.26	+0.28
Leave-one-out influence	+0.84	+0.61	+0.68	+0.50	+0.27	+0.30
Update size before clipping	+0.83	+0.58	+0.67	+0.48	+0.24	+0.27
Distance from average update	+0.72	+0.64	+0.67	+0.50	+0.27	+0.30
Cosine with average update	+0.82	+0.59	+0.70	+0.25	-0.06	+0.20
How often clipped	—	+0.48	+0.48	—	+0.27	+0.31
Label skew	-0.35	-0.21	-0.22	-0.07	+0.43	-0.18
Training loss	+0.09	+0.23	+0.18	+0.50	-0.03	+0.03

Frontier graphs: size-weighted (left), equal-weighted (right)

EuroSAT



CIFAR-10

